

Multi-aspect Diffusion Network Inference

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ABSTRACT

To learn influence relationships between nodes in a diffusion network, most existing approaches resort to precise timestamps of historical node infections. The target network is customarily assumed as an one-aspect diffusion network, with homogeneous influence relationships. Nonetheless, tracing node infection timestamps is often infeasible due to high cost, and the type of influence relationships may be heterogeneous because of the diversity of propagation media. In this work, we study how to infer a multi-aspect diffusion network with heterogeneous influence relationships, using only node infection statuses that are more readily accessible in practice. Equipped with a probabilistic generative model, we iteratively conduct a posteriori, quantitative analysis on historical diffusion results of the network, and infer the structure and strengths of homogeneous influence relationships in each aspect. Extensive experiments on both synthetic and real-world networks are conducted, and the results verify the effectiveness and efficiency of our approach.

CCS CONCEPTS

• **Computing methodologies** → **Maximum likelihood modeling**; • **Mathematics of computing** → **Computing most probable explanation**.

KEYWORDS

Multi-aspect Diffusion Network, Heterogeneous Influence Relationship, Transmission Probability

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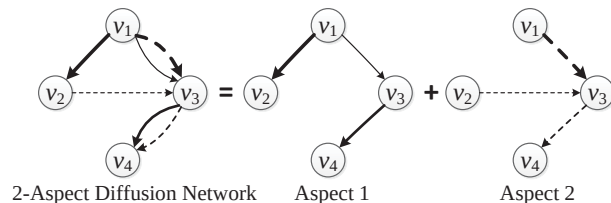


Figure 1: Example of a 2-aspect diffusion network

1 INTRODUCTION

The spread of information, viewpoints and diseases can be modeled as a stochastic process over an underlying diffusion network, where a directed edge from a parent to a child indicates that the parent can influence the child with a certain probability [4, 16]. Diffusion network inference aims to uncover such influence relationships based on spread results observed in historical diffusion processes. This problem is fundamental in applications such as information propagation [13], viral marketing [20], and epidemic prevention [32], in which the inferred influence relationships are essential for understanding the spread mechanisms and developing strategies to control future diffusion processes. Furthermore, an explicit diffusion network topology is also a foundation of influence maximization [36], community detection [23, 24], and subgraph mining [28, 34].

Most existing approaches to diffusion network inference assume that nodes infected sequentially within a short time interval are more likely to possess influence relationships. These approaches require to monitor diffusion processes constantly to obtain precise timestamps of node infections (known as cascades), although the monitoring work is often time consuming, labor/resource demanding, and sometimes dangerous in reality. Given the limitation of cascade approaches, some other approaches try to learn influence relationships between nodes without infection timestamps, using only the final infection statuses of nodes in historical diffusion processes [2, 6, 12, 15–17], since observing node infection statuses at the end of each diffusion process is more feasible than monitoring the entire diffusion process.

Both types of approaches assume that a diffusion network has only one aspect, i.e., the type of influence relationships in the network is homogeneous. However, because of the differences of propagation media, the type of influence relationships may be heterogeneous. For example, certain viruses can cause infections with similar symptoms, but they spread among humans through different types of media: air, water, or blood, forming different types of

influence relationships. Fig. 1 shows a toy diffusion network with two types of influence relationships, in which solid and dash lines refer to two types of influence relationships, and line width reflects the influence relationship strength (known as transmission probability). According to the type of influence relationships, the entire network can be decomposed into two subgraphs. Each subgraph is associated with one aspect, and the two subgraphs form a 2-aspect diffusion network. Given the fact that aspects are usually not explicitly observed, inferring a multi-aspect diffusion network is often more challenging than learning homogeneous influence relationships. So far, only one existing work [33] has partially addressed the problem of multi-aspect diffusion network inference. Under an implicit assumption that each aspect of diffusion network has the same network structure, this work tends to learn multi-aspect transmission probabilities, and is still based on cascades.

Aiming at a more general solution to diffusion network learning, we study the problem of inferring multi-aspect diffusion networks based on only the final infection statuses of nodes in historical diffusion processes. We propose an efficient and effective algorithm called LMDN (Learning Multi-aspect Diffusion Networks) for this problem. It consists of three iterative steps: (1) for the result of node infection statuses observed at the end of each historical diffusion process, estimating which aspects of the objective diffusion network have high probabilities of causing this infection status result; (2) inferring the structure of the parent-child influence relationships in each aspect based on infection status results that are regarded to be caused by this aspect; and (3) learning the most likely transmission probabilities between nodes and their parents w.r.t. hitherto inferred structure of each aspect. LMDN repeats the above three steps until the inferred structure of each aspect converges to a stable solution.

In summary, our key contributions include the following: (1) To our knowledge, this work is the first to explicitly tackle the problem of inferring a multi-aspect diffusion network from only infection status results of nodes. (2) We present a probability estimation approach to quantitatively measure the probability that an infection status result is caused by an aspect of the objective network. (3) We propose an effective approach that incorporates the probability estimation result into the structure inference and the learning of transmission probabilities for each aspect, and uncovers the objective multi-aspect diffusion network in an iterative way.

The remainder of the paper is organized as follows. We review related work in Section 2 and present a problem statement in Section 3. We then cover the proposed LMDN algorithm in Section 4, and report experimental findings in Section 5 before concluding the paper in Section 6.

2 RELATED WORK

The existing approaches to diffusion network inference can be categorized into two main groups, namely (1) one-aspect network inference, and (2) multi-aspect network inference.

2.1 One-aspect Network Inference

Most existing approaches focus on inferring one-aspect diffusion networks from cascades. A canonical strategy is adopting a convex optimization framework to find influence relationships that maximize the likelihood of the cascades [21], and utilizing techniques,

such as sequential quadratic programming [7, 25], block coordinate descent [5], stochastic and proximal gradient methods [4, 10], survival theory [9], EM algorithm [30], and sparse recovery [29], to approximate the optimal solution with reasonable time complexity. While several other approaches adopt non-convex optimization frameworks, each of them finally decouples the non-convex problem into multiple smaller convex problems [18, 26, 27]. Another effective method is transforming the problem of diffusion network inference into that of submodular optimization by constructing a likelihood function of cascades with the property of submodularity, and then adopting a greedy algorithm to achieve a near-optimal solution [8, 11]. It has been validated that with adequate amount of complete and correct cascades, homogeneous influence relationships can be accurately recovered, even using a simple inference approach [1, 3]. In addition, a few more sophisticated approaches are proposed to handle the case that cascades have partially incorrect infection timestamps [31] or miss partial snapshots of the network [14, 22, 35]. However, these approaches still require to monitoring diffusion processes to obtain cascades.

To overcome the limitation of cascade-based approaches, some existing works aim to learn homogeneous influence relationships from only the final infection statuses of nodes. A classical approach is to estimate the lifting effect of a seed node u to another infected node v [2]. This lifting effect measures the increase in the probability of v 's infection on the condition that u is infected already. A higher lifting effect indicates a higher probability that node u has an influence relationship to node v . Another class of methods aims to design a score function that takes node infection statuses as parameters, and is used to quantitatively measure the likelihood of potential structures of the objective diffusion network [6, 12, 16, 17]. As the score function is decomposable, finding the most probable structure is equivalent to finding for each node a set of most probable parent nodes with the highest likelihood in the network.

Departing from these approaches to one-aspect diffusion network inference, our LMDN algorithm is designed for multi-aspect diffusion network inference.

2.2 Multi-aspect Network Inference

Multi-aspect diffusion network inference assumes that influence relationships are heterogeneous. Different types of influence relationships form separate aspects of the diffusion network, resulting in different diffusion patterns and corresponding spread results. To the authors' knowledge, MMRate [33] is the only existing work that partially addresses multi-aspect diffusion network inference. It does not distinguish the structures of different aspects, but uncovers aspects by grouping cascades with the same diffusion pattern and inferring diverse transmission probabilities between each two nodes based on different cascade groups.

Compared with MMRate, our LMDN algorithm is able to reconstruct the different structures of the aspects, and requires only the final infection statuses of nodes, which are much more easily accessible than cascades in practice.

3 PROBLEM STATEMENT

A multi-aspect diffusion network G can be represented as a set of directed graphs: $G = \{G_1, \dots, G_\alpha\}$, where $\alpha \geq 1$ is the number

of aspects. Directed graph $G_t = \{V, E_t\}$ ($t \in \{1, \dots, \alpha\}$) is the t -th aspect of G , where $V = \{v_1, v_2, \dots, v_n\}$ is the set of n nodes in the network, and E_t is the set of directed edges (i.e., parent-child influence relationships) in the t -th aspect. Let $F_{t,i}$ denote the set of parent nodes of node v_i w.r.t. E_t , as each node has two possible infection statuses (i.e., infected status, which is denoted by 1, and uninfected status, which is denoted by 0), there are $2^{|F_{t,i}|}$ possible combinations of the infection statuses of v_i 's parent nodes, where $|F_{t,i}|$ refers to the number of nodes in set $F_{t,i}$. Let X_i and $X_{F_{t,i}}$ be the infection status variable of node v_i and the infection status variables of nodes in $F_{t,i}$, respectively, $\pi_{t,i,j}$ be the j -th possible combinations of the infection statuses of nodes in $F_{t,i}$ ($j \in \{1, \dots, 2^{|F_{t,i}|}\}$), and $\theta_{t,i,j,1}$ be the probability of v_i being infected (i.e., $X_i = 1$) under the condition that the infection statuses of nodes in $F_{t,i}$ are instantiated to the j -th possible combination (i.e., $X_{F_{t,i}} = \pi_{t,i,j}$), we refer to the probabilities $\{\theta_{t,i,j,1} \mid j \in \{1, \dots, 2^{|F_{t,i}|}\}\}$ as the transmission probabilities between $F_{t,i}$ and v_i ($\theta_{t,i,j,0} = 1 - \theta_{t,i,j,1}, \forall t, i, j$), and $\Theta_t = \{\theta_{t,i,j,1} \mid i \in \{1, \dots, n\}, j \in \{1, \dots, 2^{|F_{t,i}|}\}\}$ is the set of transmission probabilities w.r.t. E_t .

In the problem of multi-aspect diffusion network inference, the node set V is given, while the structure of each aspect, i.e., directed edge set E_t ($t \in \{1, \dots, \alpha\}$), is unknown, so are the transmission probabilities Θ_t . In this paper, we try to infer a multi-aspect diffusion network with only the final infection statuses of network nodes in a limited number of historical diffusion processes. Let $L^\ell \in \{1, \dots, \alpha\}$ be an aspect label, which indicates that the ℓ -th diffusion process takes place on the L^ℓ -th aspect, or in other words, the L^ℓ -th aspect causes the infection status result of the ℓ -th diffusion process. Then, our goal is to infer the aspect label for the infection status result observed from each diffusion process, and reconstruct the structure of each aspect with corresponding infection status results. Formally, our problem statement can be formulated as follows.

Given: a set $S = \{S^1, \dots, S^\beta\}$ of infection status results observed on an α -aspect diffusion network G in β diffusion processes, where $S^\ell = \{s_1^\ell, \dots, s_n^\ell\}$ ($\ell \in \{1, \dots, \beta\}$) is a n -dimensional vector that records the final infection status $s_i^\ell \in \{0, 1\}$ of each node $v_i \in V$ in the ℓ -th diffusion process.

Infer: the structure of each aspect of G .

4 THE LMDN ALGORITHM

To infer a multi-aspect diffusion network G with infection status results S , LMDN performs the following three steps iteratively, namely (1) estimating the aspect label of each infection status result $S^\ell \in S$, (2) inferring the structure of each aspect based on the latest aspect label estimation result, and (3) learning transmission probabilities w.r.t. hitherto inferred structure of each aspect.

In the first step, LMDN randomly initializes the value of the aspect label L^ℓ of each infection status result S^ℓ at the first iteration; in subsequent iterations, it estimates the probability distribution of the value of each L^ℓ , based on infection status results S , the latest inferred structure of each aspect, and the corresponding transmission probabilities.

In the second step, LMDN leverages an existing approach to one-aspect diffusion inference to infer the structure of each aspect with infection status results that have corresponding aspect label. At the first iteration, LMDN utilizes initialized aspect labels directly; in

subsequent iterations, it samples the value of each aspect label L^ℓ based on the latest estimated probability distribution of L^ℓ 's value.

In the third step, LMDN performs an EM-like approach to approximate the optimal values of transmission probabilities w.r.t. the latest inferred structure of each aspect. The EM-like approach adopts a carefully designed expectation function, from which the optimal updating result for current values of transmission probabilities can be theoretically deduced. EM-like approach iteratively updates the values of transmission probabilities until convergence.

The details of the above three steps, as well as a complexity analysis for LMDN, are provided in what follows.

4.1 Aspect Label Estimation

Given the α -aspect diffusion network $G^{(T)} = \{V, E^{(T)}\}$ as well as corresponding transmission probabilities $\Theta^{(T)} = \{\Theta_1^{(T)}, \dots, \Theta_\alpha^{(T)}\}$ inferred by LMDN after T iterations, where $E^{(T)} = \{E_1^{(T)}, \dots, E_\alpha^{(T)}\}$, the posteriori probability that the ℓ -th infection status result S^ℓ is caused by the t -th aspect of $G^{(T)}$ can be calculated as follows.

$$\begin{aligned} & P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)}) \\ &= P(L^\ell = t \mid S^\ell, E^{(T)}, \Theta^{(T)}) \\ &= \frac{P(L^\ell = t, S^\ell \mid E^{(T)}, \Theta^{(T)})}{\sum_{k=1}^{\alpha} P(L^\ell = k, S^\ell \mid E^{(T)}, \Theta^{(T)})} \\ &= \frac{P(S^\ell \mid L^\ell = t, E^{(T)}, \Theta^{(T)})P(L^\ell = t)}{\sum_{k=1}^{\alpha} P(S^\ell \mid L^\ell = k, E^{(T)}, \Theta^{(T)})P(L^\ell = k)}, \end{aligned} \quad (1)$$

where $P(L^\ell = t)$ refers to a prior probability that the ℓ -th diffusion process takes place on the t -th aspect of the objective α -aspect diffusion network.

As there is no prior knowledge on the value of $P(L^\ell = t)$, we are indifferent to regard every possible value of $P(L^\ell = t)$. In other words, we assume that $P(L^\ell = t) = \frac{1}{\alpha}, \forall t$. Then, the posteriori probability $P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})$ can be estimated as follows.

$$\begin{aligned} & P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)}) \\ &= \frac{P(S^\ell \mid L^\ell = t, E^{(T)}, \Theta^{(T)}) \frac{1}{\alpha}}{\sum_{k=1}^{\alpha} P(S^\ell \mid L^\ell = k, E^{(T)}, \Theta^{(T)}) \frac{1}{\alpha}} \\ &= \frac{P(S^\ell \mid L^\ell = t, E^{(T)}, \Theta^{(T)})}{\sum_{k=1}^{\alpha} P(S^\ell \mid L^\ell = k, E^{(T)}, \Theta^{(T)})}, \end{aligned} \quad (2)$$

where

$$\begin{aligned} & P(S^\ell \mid L^\ell = t, E^{(T)}, \Theta^{(T)}) \\ &= P(S^\ell \mid E_t^{(T)}, \Theta_t^{(T)}) \\ &= \prod_{i=1}^n P(X_i = s_i^\ell \mid X_{F_{t,i}} = \pi_{t,i,j(\ell)}^{(T)}, E_t^{(T)}, \Theta_t^{(T)}) \\ &= \prod_{i=1}^n \theta_{t,i,j(\ell),s_i^\ell}^{(T)} \end{aligned} \quad (3)$$

and $\pi_{t,i,j(\ell)}^{(T)}$ denotes the infection statuses of v_i 's parent nodes w.r.t. current $E_t^{(T)}$ in the ℓ -th diffusion process, $\theta_{t,i,j(\ell),s_i^\ell}^{(T)}$ is the probability of v_i 's infection status being s_i^ℓ on the condition that

the infection statuses of v_i 's parent nodes w.r.t. current $E_t^{(T)}$ are instantiated to $\pi_{t,i,j(\ell)}^{(T)}$.

A greater value of $P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})$ indicates that the ℓ -th infection status result S^ℓ is more likely to be caused by the t -th aspect of $G^{(T)}$.

4.2 Structure Inference

After estimating probabilities $\{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\}$ based on $E^{(T)}$, $\Theta^{(T)}$, and infection status results S , we can update the result of structure inference by finding a directed edge set $E^{(T+1)}$ that satisfies the following requirement.

$$E^{(T+1)} = \arg \max_E P(E \mid S, \{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\}), \quad (4)$$

where $\ell \in \{1, \dots, \beta\}$ and $t \in \{1, \dots, \alpha\}$.

Nonetheless, as $P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})$ describes the posteriori probability of $L^\ell = t$ rather than the exact aspect label of S^ℓ , it is difficult to calculate the probability $P(E \mid S, \{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\})$ directly. Aiming at an easier estimation for $P(E \mid S, \{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\})$, we repeatedly sample the value of each aspect label L^ℓ from $\{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\}$ to obtain a set of samples $\{\hat{L}_1^\ell, \dots, \hat{L}_m^\ell\}$ for L^ℓ , where m is the times of sampling. Then, we obtain a set of labeled infection status results: $\{(S, \hat{L}_r) \mid r \in \{1, \dots, m\}\}$, where $(S, \hat{L}_r) = \{(S^\ell, \hat{L}_r^\ell) \mid \ell \in \{1, \dots, \beta\}\}$. If the sampling is sufficient (i.e., m is great enough), then

$$\begin{aligned} P(E \mid S, \{P(L^\ell = t \mid S, E^{(T)}, \Theta^{(T)})\}) \\ \approx P(E \mid (S, \hat{L}_1), \dots, (S, \hat{L}_m)). \end{aligned} \quad (5)$$

Given this, we can reformulate Eq. (4) as follows.

$$E^{(T+1)} = \arg \max_E P(E \mid (S, \hat{L}_1), \dots, (S, \hat{L}_m)). \quad (6)$$

Let S_t be the collection of labeled infection status results with aspect label t in $\{(S, \hat{L}_r) \mid r \in \{1, \dots, m\}\}$, i.e.,

$$S_t = \{S^\ell \mid \hat{L}_r^\ell = t, \ell \in \{1, \dots, \beta\}, r \in \{1, \dots, m\}\}. \quad (7)$$

Note that S^ℓ may appear in S_t more than one time since there may be more than one $r \in \{1, \dots, m\}$ such that $\hat{L}_r^\ell = t$.

With the S_t collected in the T -th iteration, the optimization problem in Eq. (6) can be divided into α optimization problems as follows.

$$E_t^{(T+1)} = \arg \max_E P(E \mid S_t), \quad t \in \{1, \dots, \alpha\}. \quad (8)$$

As each of the above optimization problem is equivalent to learning an one-aspect diffusion network, one can utilize existing approaches to one-aspect diffusion network inference, such as SIDN [16], to infer each $E_t^{(T+1)}$.

4.3 Learning Transmission Probabilities

With the inferred $E^{(T+1)}$, we can learn the most likely transmission probabilities $\Theta^{(T+1)}$ by solving the following optimization problem.

$$\Theta^{(T+1)} = \arg \max_{\Theta} P(\Theta \mid S, E^{(T+1)}). \quad (9)$$

Given $E^{(T+1)}$, the parent nodes of a node $v_i \in V$ and the possible combinations of v_i 's parent nodes in each aspect of $G^{(T+1)}$ are fixed,

so are the elements in $\Theta^{(T+1)}$. What we need to do is to find each element $\theta_{t,i,j,1}^{(T+1)} \in \Theta^{(T+1)}$ an optimal value such that the likelihood of observed infection status results S is maximized.

Let $H(E^{(T+1)})$ denote the constrained search space of $\Theta^{(T+1)}$ w.r.t. the given $E^{(T+1)}$, the problem of learning $\Theta^{(T+1)}$ can be reformulated as

$$\Theta^{(T+1)} = \arg \max_{\Theta \in H(E^{(T+1)})} P(S \mid \Theta). \quad (10)$$

Inspired by the EM algorithm, we propose to update Θ iteratively, and define an expectation function Q as follows.

$$\begin{aligned} Q(\Theta, \Theta^{[\tau]}) &= E_{L \sim P(L \mid S, \Theta^{[\tau]})} [\log P(S, L \mid \Theta) \mid S, \Theta^{[\tau]}] \\ &= E_{L \sim P(L \mid S, \Theta^{[\tau]})} [\log \prod_{\ell=1}^{\beta} P(S^\ell, L^\ell \mid \Theta) \mid S^\ell, \Theta^{[\tau]}] \\ &= E_{L \sim P(L \mid S, \Theta^{[\tau]})} [\sum_{\ell=1}^{\beta} \log P(S^\ell, L^\ell \mid \Theta) \mid S^\ell, \Theta^{[\tau]}] \\ &= \sum_{\ell=1}^{\beta} E_{L \sim P(L \mid S, \Theta^{[\tau]})} [\log P(S^\ell, L^\ell \mid \Theta) \mid S^\ell, \Theta^{[\tau]}] \\ &= \sum_{\ell=1}^{\beta} \sum_{t=1}^{\alpha} P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) \log P(S^\ell, L^\ell = t \mid \Theta). \end{aligned} \quad (11)$$

where $\Theta^{[\tau]}$ refers to the updated Θ after τ iterations, and $E[\cdot]$ refers to the expectation value of a variable.

Function Q has a nice theoretical property as follows.

THEOREM 1. *If $Q(\Theta, \Theta^{[\tau]}) > Q(\Theta^{[\tau]}, \Theta^{[\tau]})$, then relationship $P(S \mid \Theta) > P(S \mid \Theta^{[\tau]})$ holds.*

Proof: The relationship

$$\begin{aligned} &\log P(S \mid \Theta) - \log P(S \mid \Theta^{[\tau]}) \\ &= \sum_{\ell=1}^{\beta} \log P(S^\ell \mid \Theta) - \sum_{\ell=1}^{\beta} \log P(S^\ell \mid \Theta^{[\tau]}) \\ &= \sum_{\ell=1}^{\beta} \log \sum_{t=1}^{\alpha} \frac{P(S^\ell, L^\ell = t \mid \Theta)}{P(S^\ell \mid \Theta^{[\tau]})} \frac{P(L^\ell = t \mid \Theta^{[\tau]}, S^\ell)}{P(L^\ell = t \mid \Theta^{[\tau]}, S^\ell)} \\ &= \sum_{\ell=1}^{\beta} \log \sum_{t=1}^{\alpha} P(L^\ell = t \mid \Theta^{[\tau]}, S^\ell) \frac{P(S^\ell, L^\ell = t \mid \Theta)}{P(S^\ell, L^\ell = t \mid \Theta^{[\tau]})} \\ &\geq \sum_{\ell=1}^{\beta} \sum_{t=1}^{\alpha} P(L^\ell = t \mid \Theta^{[\tau]}, S^\ell) \log \frac{P(S^\ell, L^\ell = t \mid \Theta)}{P(S^\ell, L^\ell = t \mid \Theta^{[\tau]})} \\ &= Q(\Theta, \Theta^{[\tau]}) - Q(\Theta^{[\tau]}, \Theta^{[\tau]}) \end{aligned}$$

holds, where the inequality is derived from Jensens inequality. Therefore, the theorem is correct. ■

According to the above Theorem 1, one can learn the optimal $\Theta^{[\tau+1]}$ by finding a Θ from $H(E^{(T+1)})$ such that the value of $Q(\Theta, \Theta^{[\tau]})$ is maximized, i.e.,

$$\Theta^{[\tau+1]} = \arg \max_{\Theta \in H(E^{(T+1)})} Q(\Theta, \Theta^{[\tau]}). \quad (12)$$

According to Eqs. (2) & (3), the probability $P(L^\ell = t \mid S^\ell, \Theta^{[\tau]})$ in Eq. (11) can be calculated as

$$P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) = \frac{\prod_{i=1}^n \theta_{t,i,j(\ell),s_i^\ell}^{[\tau]}}{\sum_{k=1}^\alpha \prod_{i=1}^n \theta_{k,i,j(\ell),s_i^\ell}^{[\tau]}}. \quad (13)$$

In addition, $\log P(S^\ell, L^\ell = t \mid \Theta) = \log P(S^\ell \mid L^\ell = t, \Theta)P(L^\ell = t \mid \Theta)$, where $P(L^\ell = t \mid \Theta)$ represents probability $P(L^\ell = t)$ under condition that Θ (i.e., transmission probabilities) is known. The condition doesn't include observation data S^ℓ . Without S^ℓ , $P(L^\ell = t)$ is assumed to be independent to Θ , i.e., $P(L^\ell = t \mid \Theta) = P(L^\ell = t)$. Then,

$$\begin{aligned} & \log P(S^\ell, L^\ell = t \mid \Theta) \\ &= \log P(S^\ell \mid L^\ell = t, \Theta)P(L^\ell = t) \\ &= \log \frac{1}{\alpha} \prod_{i=1}^n \theta_{t,i,j(\ell),s_i^\ell} \\ &= \sum_{i=1}^n \log \theta_{t,i,j(\ell),s_i^\ell} - \log \alpha, \end{aligned} \quad (14)$$

and the expectation function Q defined in Eq. (11) can be reformulated as follows.

$$\begin{aligned} & Q(\Theta, \Theta^{[\tau]}) \\ &= \sum_{\ell=1}^\beta \sum_{t=1}^\alpha P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) \left(\sum_{i=1}^n \log \theta_{t,i,j(\ell),s_i^\ell} - \log \alpha \right) \\ &= \sum_{\ell=1}^\alpha \sum_{i=1}^n \sum_{j=1}^{2^{|F_{t,i}^{(T+1)}|}} (h_{t,i,j} - c_{t,i,j}), \end{aligned} \quad (15)$$

where $F_{t,i}^{(T+1)}$ refers to the set of parent nodes of node v_i determined by $E_t^{(T+1)}$, and

$$\begin{aligned} h_{t,i,j} &= a_{t,i,j} \log(1 - \theta_{t,i,j,1}) + b_{t,i,j} \log \theta_{t,i,j,1}, \\ a_{t,i,j} &= \sum_{\ell=1}^\beta P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) (1 - s_i^\ell) I(j(\ell) = j), \\ b_{t,i,j} &= \sum_{\ell=1}^\beta P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) s_i^\ell I(j(\ell) = j), \\ c_{t,i,j} &= \sum_{\ell=1}^\beta P(L^\ell = t \mid S^\ell, \Theta^{[\tau]}) \log \alpha. \end{aligned} \quad (16)$$

In the above equations, $I(\cdot)$ is an indicator function.

The following theorem can help us find an optimal Θ that maximizes $Q(\Theta, \Theta^{[\tau]})$.

CLAIM 1. Let $0 < \theta < 1$, $a > 0$ and $b > 0$, then function $f(\theta) = a \log \theta + b \log(1 - \theta)$ reaches its maximum value at $\theta = \frac{a}{a+b}$.

Proof: The derivative of $f(\theta)$ is $f'(\theta) = \frac{a}{\theta} - \frac{b}{1-\theta}$, based on which we have that for $0 < \theta < \frac{a}{a+b}$, $f'(\theta) > 0$ holds, and for $\frac{a}{a+b} < \theta < 1$, relationship $f'(\theta) < 0$ holds. Hence, $f(\theta)$ reaches its maximum value at $\theta = \frac{a}{a+b}$, and the theorem is correct. ■

As $c_{t,i,j}$ is a constant and is not a coefficient of any variable, to maximize $Q(\Theta, \Theta^{[\tau]})$, we should maximize the value of each $h_{t,i,j}$. According to Claim 1, each $h_{t,i,j}$ reaches its maximum value at

$b_{t,i,j} / (a_{t,i,j} + b_{t,i,j})$. In other words, the optimal $\theta_{t,i,j,1}^{(\tau+1)} \in \Theta^{(T+1)}$ should be

$$\theta_{t,i,j,1}^{(\tau+1)} = \frac{b_{t,i,j}}{a_{t,i,j} + b_{t,i,j}}. \quad (17)$$

Given the discussion above, the optimal values of $\Theta^{(T+1)}$ w.r.t. given $E^{(T+1)}$ can be iteratively approximated by repeatedly using Eq. (12).

4.4 Complexity Analysis

In aspect label estimation, the most computationally expensive process is estimating the probability in Eq. (3) for each infection status result S^ℓ and each aspect t , the time complexity of this process is about $O(\alpha \beta n^2)$, where n is the number of nodes in objective diffusion network, α is the number of aspects, β is number of infection status results, and η is an upper bound of the number of parent nodes for each node in each aspect ($\eta \ll n$). Therefore, when the objective networks have dense structures, aspect label estimation will become computationally expensive due to a large η .

In structure inference, performing SIDN on labeled infection status results to infer the structure of each aspect takes about $O(m \beta n^2)$ time, where m is sampling times for the value of each aspect label. Then, inferring the structures of α aspects takes $O(m \alpha \beta n^2)$ time.

Learning transmission probabilities is carried out in an iterative way. In each learning iteration, the most computationally expensive process is calculating Eq. (13) and $a_{t,i,j}$, $b_{t,i,j}$ in Eq. (16), which takes $O(\alpha \beta n^2)$ time. Let τ be the number of all learning iterations, learning transmission probabilities takes $O(\tau \alpha \beta n^2)$ time in all.

In summary, the time complexity of LMDN is $O(m T \alpha \beta n^2 + T \tau \alpha \beta n^2)$, where T refers to the number of iterations of LMDN.

5 EXPERIMENTAL EVALUATION

In this section, we first introduce the experimental setup, and then verify the effectiveness and efficiency of our LMDN algorithm on synthetic and real-world networks. We investigate the effects of the number of aspects, diffusion network size, initial infection ratio, the amount of diffusion processes, and the iterations of LMDN, on the accuracy and running time of LMDN. All algorithms in the experiments are implemented in Python, running on a desktop PC with Intel Core i9-9900k CPU at 3.60GHz and 32GB RAM. The source code of LMDN and the data used in the experiments are available at <https://github.com/Diffusion-Network-Inference/LMDN>.

5.1 Experimental Setup

Network. We adopt LFR benchmark graphs [19] as the synthetic one-aspect networks. By setting different generation parameters, such as the number n of nodes and the average degree of each node, we generate a series of LFR benchmark graphs with properties summarized in Table 1. Then, we generate nine synthetic multi-aspect diffusion networks, each of which consists of α LFR benchmark graphs that have the same number n of nodes, but different edges and different average degrees. One LFR benchmark graph forms one aspect. Table 2 summarizes the properties of these synthetic multi-aspect diffusion networks. In addition, we also generate two multi-aspect diffusion networks with the following two real-world microblogging networks [33], i.e., (1) DUNF, which contains 750

Table 1: Properties of LFR benchmark graphs

Graphs	Number n of Nodes	Average Degree
G1	200	2
G2-G6	100, 150, 200, 250, 300	3
G7-G11	100, 150, 200, 250, 300	4
G12-G16	100, 150, 200, 250, 300	5
G17	200	6

Table 2: Properties of synthetic α -aspect networks

α -Aspect Network	α	Aspects	n
N1	3	G2, G7, G12	100
N2	3	G3, G8, G13	150
N3	3	G4, G9, G14	200
N4	3	G5, G10, G15	250
N5	3	G6, G11, G16	300
N6	1	G9	200
N7	2	G4, G9	200
N8	4	G1, G4, G9, G14	200
N9	5	G1, G4, G9, G14, G17	200

users and 2974 following relationships, and (2) DPU, which contains 1038 users and 11385 following relationships. Specifically, we first obtain three sub-graphs of both DUNF and DPU by randomly sampling the edges of these two networks at a ratio of 1/3, and then generate two 3-aspect diffusion networks by using each sub-graph of DUNF and DPU as one aspect. We refer to these two 3-aspect diffusion networks as 3-DUNF and 3-DPU, respectively. The two microblogging networks and a similar multi-aspect diffusion network generation method are used in the existing study of multi-aspect diffusion network inference [33].

Infection Status Data. The infection status results $S = \{S_1, \dots, S_\beta\}$ can be generated by simulating β times of diffusion processes on each multi-aspect network with randomly selected initially infected nodes in each simulation. Each diffusion process takes place on only one aspect of the multi-aspect network. In each diffusion process, each infected node tries to infect its uninfected child nodes with a certain probability, which subjects to a Gaussian distribution with a mean of 0.3 and a standard deviation of 0.05, so that about 95% of values of the probability are within a range from 0.2 to 0.4. Besides infection status results S , the corresponding cascades (i.e., infection timestamps) are also recorded for cascade-based tested algorithms in the experiments. Similar generation methods for infection status data and cascades are commonly used in existing studies [2, 6, 8, 12, 16, 17, 33].

Performance Criterion. We evaluate the performance of LMDN in terms of the accuracy of structure inference. For each aspect of the objective multi-aspect diffusion network, the accuracy of structure inference for this aspect can be measured by the F-score (i.e., the harmonic mean of precision and recall) of the inferred directed edges, which can be calculated as $F\text{-score} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$, where $\text{precision} = \frac{N_{TP}}{N_{TP} + N_{FP}}$, $\text{recall} = \frac{N_{TP}}{N_{TP} + N_{FN}}$, N_{TP} is the number of true positives, i.e., the true edges which are correctly inferred by the algorithm; N_{FP} is the number of false positives, i.e., the wrong inferred edges which are not in the real network; and N_{FN}

is the number of false negatives, i.e., the true edges which are not correctly inferred by the algorithm. We report the average F-score on all aspect of a multi-aspect diffusion network as the accuracy performance of structure inference on this multi-aspect diffusion network. A greater value of F-score indicates a more accurate result of structure inference.

Benchmark Algorithms. We compare LMDN with (1) MMRate [33], which is, to the best of our knowledge, the only existing approach to multi-aspect diffusion network inference, (2) NetRate [7], which is a classical cascade-based approach to one-aspect diffusion network inference, and (3) SIDN [16], which has a high-performance infection status-based approach to one-aspect diffusion network inference and is also adopted in LMDN to infer the structure of each aspect with infection status results labeled by LMDN. For our LMDN algorithm, the number m of aspect label sampling round is set to 6, the maximum number T of iterations is set to 15, and the stop condition for the iterative updates of transmission probabilities is that the variation of each transmission probability is less than 0.01. Since MMRate implicitly assumes each aspect has the same structure and tries to infer the pairwise transmission probabilities between each two nodes in every aspect from cascades, we give it a privilege in the accuracy performance comparison of structure inference, i.e., by calculating the F-score of edges whose transmission probabilities are greater than a threshold, we use different thresholds to find a highest F-score on each aspect, and then report the mean of these highest F-scores as MMRate’s final accuracy performance on structure inference. A similar privilege is also given to NetRate. Since NetRate and SIDN are approaches to one-aspect diffusion network inference, when we execute NetRate and SIDN on a multi-aspect diffusion network, we treat the objective network as an one-aspect network regardless of the type of influence relationships. If a directed edge inferred by NetRate or SIDN exists in any aspect of the objective multi-aspect diffusion network, we regard it as a true positive.

5.2 Effect of The Number of Aspects

To study the effect of the number of aspects on algorithms’ performance, we compare LMDN with MMRate, NetRate and SIDN on five synthetic multi-aspect networks, i.e., N6, N7, N3, N8, and N9, which have the same network size ($n = 200$) but different numbers of aspects (α varies from 1 to 5). We simulate 200 diffusion processes on each aspect, and thus observing 200, 400, 600, 800, 1000 infection status results/cascades on these five multi-aspect networks, respectively. In each of these diffusion processes, $0.15n$ nodes are randomly selected as the initially infected nodes.

Fig. 2 reports the accuracy performance (standard deviation is showed by shaded area) and running time of each algorithm, from which we can have the following observations.

(1) LMDN has the highest accuracy, and shows an insensitivity to the number of aspects. This is because LMDN can always conduct an accurate aspect label estimation, which enables an accurate structure inference for each aspect.

(2) The increase of the number of aspects tends to cause performance degradation for MMRate, NetRate, and SIDN. For MMRate, this is because it is prone to dividing aspect labels into one or two main groups, thus making more mistakes on aspect label estimation,

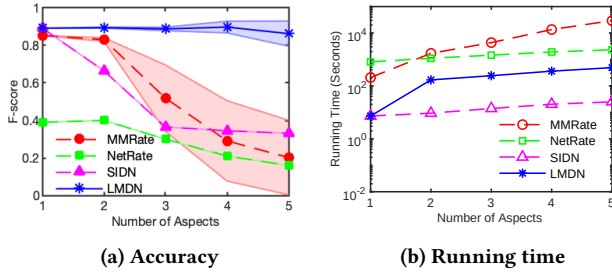


Figure 2: Effect of the number of aspects

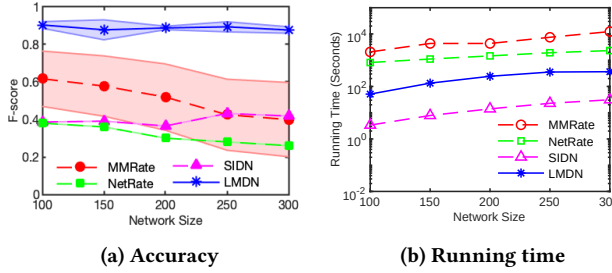


Figure 3: Effect of diffusion network size

especially when there are more than two aspects. For NetRate and SIDN, this is because they treat the objective multi-aspect network as an one-aspect network. Then, more aspects in objective network often result in a more complex one-aspect network for them. This observation indicates that straightforward treating a multi-aspect network as an one-aspect network is not a very suitable solution for multi-aspect diffusion network inference.

(3) The running time of each algorithm increases with the number of aspects. LMDN is an order of magnitude faster than MMRate and NetRate, and SIDN executes the fastest (but can not really uncover multi-aspect diffusion networks).

5.3 Effect of Diffusion Network Size

To study the effect of diffusion network size on algorithms' performance, we compare LMDN with MMRate, NetRate and SIDN on five synthetic 3-aspect networks, i.e., N1, N2, N3, N4, and N5, which have different network size (n varies from 100 to 300). For each 3-aspect network, we simulate 150, 200, 250 diffusion processes on its three aspects, respectively, thus observing 600 infection status results/cascades. In each of these diffusion processes, $0.15n$ nodes are randomly selected as the initially infected nodes.

Fig. 3 illustrates the performance comparison results, from which we can observe that LMDN outperforms MMRate, NetRate, and SIDN in terms of accuracy, and we can also have the following observations.

(1) The accuracy of LMDN and SIDN is reasonably insensitive to diffusion network size. SIDN's insensitivity to diffusion network size has also been verified in existing study [16]. This is one of reasons for why LMDN adopts SIDN to infer the structure of each aspect with labeled infection status results. When LMDN is able to estimate aspect labels accurately, SIDN helps LMDN to achieve a robust performance on the structure inference of each aspect.

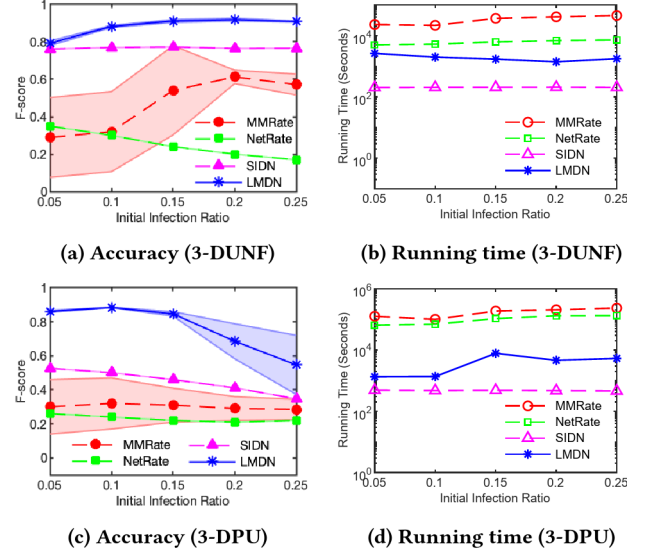


Figure 4: Effect of initial infection ratio

(2) Larger network sizes tend to degrade the accuracy of MMRate and NetRate. This is because the two algorithms infer influence relationships by checking whether the infections of two nodes are often within a time interval in historical diffusion processes. Nonetheless, a few nodes may be infected at nearly the same time, even though they have no direct influence relationship. In a larger network, more nodes are likely to be involved in each diffusion process, causing more aforementioned phenomena, which will result in more false positives in the inference results of MMRate and NetRate.

(3) The running time of each algorithm increases with the diffusion network size. LMDN has better running time performance than MMRate and NetRate, and SIDN is fastest.

In addition, based on extensive testing on larger networks, we have found that the running time of MMRate and NetRate rapidly increase with the growth of network size, and soon exceeds acceptable levels. Given this fact, we have selected networks with sizes varying from 100 to 1038. Most existing related work adopt similar network sizes.

5.4 Effect of Initial Infection Ratio

The ratio of initially infected nodes may affect the number of final infected nodes in a diffusion process. To study the effect of initial infection ratio on algorithms' performance, we compare LMDN with MMRate, NetRate and SIDN on two 3-aspect diffusion networks, i.e., 3-DUNF and 3-DPU, which are generated based on real-world microblogging networks. We vary the initial infection ratio from 0.05 to 0.25 with an interval of 0.05. With each initial infection ratio, we simulate 150, 200, 250 diffusion processes on the three aspects of each 3-aspect diffusion network.

Fig. 4 illustrates the performance comparison results, from which we can observe that LMDN performs better than other tested algorithms in terms of accuracy, and we can also have the following observations.

(1) A greater initial infection ratio tends to help LMDN to achieve a higher accuracy, until the initial infection ratio exceeds a certain

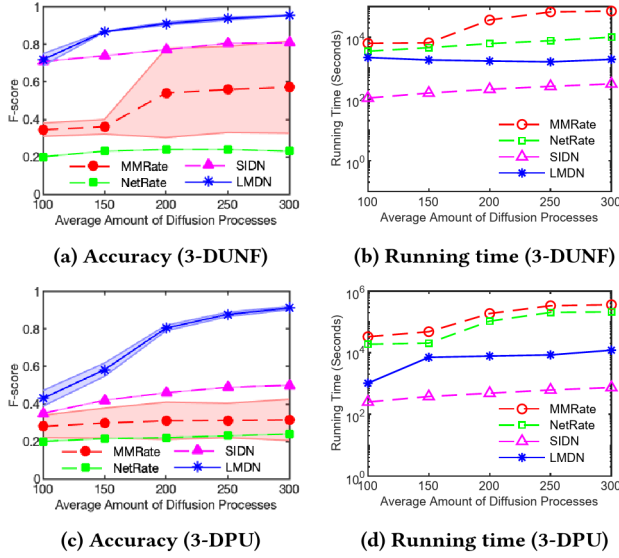


Figure 5: Effect of amount of diffusion processes

threshold. This is because in each diffusion process, the final infected nodes caused by a small ratio of initially infected nodes are insufficient to cover all nodes in objective network, so that the infection data is insufficient to help LMDN to uncover all influence relationships between the nodes. When the initial infection ratio is great enough, the final infected nodes will be sufficient to cover the majority parts of objective network, and the infection data will be sufficient to support LMDN to approximate its best accuracy performance. Nonetheless, when the initial infection ratio becomes too great, more node pairs that have no direct influence relationships may be infected simultaneously in different diffusion processes, resulting in fake statistical correlations of node infections, which may degrade the accuracy of LMDN.

(2) LMDN has better running time performance than MMRate and NetRate, while SIDN is fastest.

5.5 Effect of Diffusion Process Amount

To study the effect of the amount of diffusion processes on algorithms' performance, we compare LMDN with MMRate, NetRate and SIDN on two 3-aspect diffusion networks, i.e., 3-DPU and 3-DUNF, with different amount of diffusion processes. Towards this, for each 3-aspect diffusion network, we simulate $\beta_{ave} - 50$, β_{ave} , $\beta_{ave} + 50$ diffusion processes on its three aspects, respectively, where β_{ave} refers to the average amount of simulated diffusion processes on each aspect. We vary the average amount β_{ave} of simulated diffusion processes from 100 to 300 with an interval of 50. In each of these diffusion processes, $0.15n$ nodes are randomly selected as the initial infected nodes.

Fig. 5 illustrates the performance comparison results, from which we can have the following observations.

(1) A greater amount of diffusion processes often helps the algorithms achieve more accurate results. The reason behind is that more diffusion processes tend to involve more nodes, and activate more influence relationships (i.e., infections spread through these

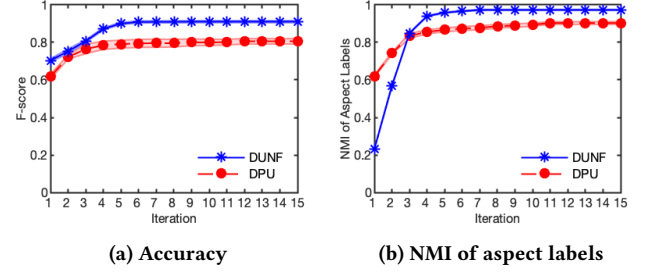


Figure 6: Effect of iterations of LMDN

influence relationships), which enable the algorithms to learn a more complete structure for the objective diffusion network.

(2) Compared with the other algorithms, LMDN achieves significantly better accuracy in all but one setting, i.e., $\beta_{ave} = 100$. This is because when $\beta_{ave} = 100$, there are only 50 diffusion processes taking place on one of the three aspects, which are insufficient to activate enough influence relationships, thus degrading the accuracy of LMDN on this aspect. In contrast, when we measure the accuracy of SIDN, we incorporate all aspects. For example, when a directed edge exists in multiple aspects, even if it is activated in only one aspect but captured by SIDN, it will be regarded as a true positive on all of these multiple aspects. This privilege tends to increase the accuracy score of SIDN.

(3) LMDN shows a significant running time advantage over MMRate and NetRate, while SIDN is fastest.

5.6 Effect of Iterations of LMDN

To study the effect of iterations on the performance of LMDN, we execute LMDN on 3-DPU and 3-DUNF and report the accuracy of its structure inference result at each iteration. For each network, we simulate 150, 200, 250 diffusion processes on its three aspects, respectively, with an initial infection ratio of 0.15 in each simulation.

Fig. 6 (a) illustrates F-score of LMDN at the end of each iteration, from which we can observe that the accuracy of LMDN can be improved with more iterations, and shows a reasonably fast convergence property.

One of the most important reasons behind this observation is that LMDN can accurately estimate the aspect labels of infection status results. To verify this point, Fig. 6 (b) illustrates the accuracy of estimated aspect labels in each iteration in terms of NMI (Normalized Mutual Information). A greater NMI value indicates a higher accuracy. From Fig. 6 (b), we can observe that the accuracy of estimated aspect labels can also be improved with more iterations, and converges quickly.

6 CONCLUSION

In this paper, we have investigated the problem of inferring a multi-aspect diffusion network with the infection status results of nodes. Towards this, we have proposed an efficient approach, LMDN, to solve this problem by iteratively estimating the aspect label of each infection status result, inferring the structure of each aspect based on aspect label estimation result, and learning transmission probabilities w.r.t. the inferred structure of each aspect. Experimental results on both synthetic and real-world networks have verified the effectiveness and efficiency of LMDN.

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